JÖNKÖPING UNIVERSITY
School of Engineering

TARI29 Artificial Intelligence, 7.5 credits, Autumn 2024

Practice Exam

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Allowed aids:

- VS Code and SWI-Prolog installed at the computers in the exam room
- SWI-Prolog documentation provided as PDF
- The logical equivalence sheet provided as PDF
- Other material provided as PDF

However, you are not allowed to use any other form form of assistance. Specifically, it is of course strictly prohibited to collaborate with other students or use online resources during the exam.

Instructions for answering questions:

- Each question will provide a task description and the maximum amount of points that can be awarded for a correct answer.
- Type in your answers directly in the Inspera system.
- If a question mentions that you can submit a drawing or other type of writing on paper, you can get paper sheets from the personnel in the exam room. Then you need to hand in your sheets to the personnel. The paper sheets will be delivered to the examiners after the exam. The paper sheets will be delivered to the examiners after the exam. If you write on paper, **write in legible (clear) handwriting** so that your answer is well readable.
- For Prolog problems, write and debug your code in the VS Code environment. Then copy the Prolog code of your solution from VS Code and paste it directly in the text field of the corresponding question.

Grading:
The maximum total score for the written examination is 70 points, and the grading is:

3	28–41.5	points
4	42–55.5	points
5	56–70	points

An example of distribution of points among questions:

Question:	Search	Game Theory	MDP	Logic	Prolog	Total
Points:	12	12	9	14	23	70

Good Luck!

1. (12 points) **Search**

We have studied the 8-puzzle in the course when discussing A*. Below are four statements about solving the 8-puzzle with uninformed search. For each statement, state whether it is true or false, and provide a very short motivation for your answer. Each statement is worth 4 points.

- Breadth-first search will always find a solution.
- Using depth-first search, the first solution found will always be optimal, i.e., have the minimal number of moves.
- Using iterative deepening search, the first solution found will always be optimal, i.e., have the minimal number of moves.

Marking Notes:

- Breadth-first search will always find a solution. TRUE. Breadth-first always expands the shallowest node. So, after a finite number of steps, the goal node will be discovered.
- Using depth-first search, the first solution found will always be optimal, i.e., have the minimal number of moves. FALSE. Once a solution is found, there could be shorter solutions in the part of the tree not explored.
- Using iterative deepening search, the first solution found will always be optimal, i.e., have the minimal number of moves. TRUE. While using depth-first search as the main search strategy, this is done one level at the time, i.e., always exhausting all n-move sequences before looking for any (n+1)-move sequence.

2. Game Theory

(a) (5 points) Consider the payoff matrix below for a game in normal form.

		Player B		
		X	**Y**	**Z**
	X	**4,6**	**3,1**	**1,5**
Player A	**Y**	**2,2**	**1,4**	**1,4**
	Z	**3,3**	**0,0**	**2,6**

Are there any pure Nash Equilibria or Pareto Optimal outcomes in the game? If so, what are they (and why)?

Marking Notes:
In a Nash Equilibrium, no player wants to change to another possible action since the payoff will be smaller.

(**X=4**,**X=6**) is a Nash Equilibrium since **Player A** doesn't want to choose action **Y=2** or **Z=3**, since action **X=4** yields a higher payoff, and **Player B** doesn't want to choose action **Y=1** or **Z=5**, since action **X=6** yields a higher payoff.

(**Z=2**,**Z=6**) is a Nash Equilibrium since **Player A** doesn't want to choose action **X=1** or **Y=1**, since action **Z=2** yields a higher payoff, and **Player B** doesn't want to choose action **X=3** or **Y=0**, since action **Z=6** yields a higher payoff.

This is not true for any other action profile, hence, (**X=4**,**X=6**) and (**Z=2**,**Z=6**) are the only Nash Equilibria in the game.

Pareto optimality is defined as:

- If an outcome o is at least as good for each agent as another outcome o', where at least one agent strictly prefers o over o', then o Pareto dominates o'.
- An outcome o^* is Pareto optimal if there isn't any other outcome that Pareto dominates o^*.

(**X=4**,**X=6**) is a Pareto Optimal outcome since it is not Pareto dominated by any other outcome (for example, no other outcome has a higher value for **Player B**). Furthermore, (**Z=2**,**Z=6**) Pareto dominates all other outcomes (since both values for **Player A** and **Player B** are at least as high than all other outcomes, where at least one value is strictly higher).

(**X=4**,**X=6**) is the only Pareto Optimal outcome in the game.

Marking:

- -1p for each incorrect or missing Nash Equilibria. ($\times 2$)
- -0.5p for insufficient justification of Nash Equilibria. ($\times 2$)
- -0.5p for each incorrect or missing Pareto Optimal outcome. ($\times 2$)
- -0.5p for insufficient justification of Pareto Optimal outcomes. ($\times 2$)

(b) (3 points) Let us assume the following scenario. Husqvarna and STIHL, two leading tool manufacturers, produce an identical robotic lawn mower. They can produce high or low quantities. Based on their decision, the following cases may occur:

- if they cooperate and agree to produce at low levels, then limited supply will translate into a high price for the product on the market and substantial profits of 40 million SEK for both companies
- if they defect and produce at high levels, the market will be swamped and result in a low price for the product and consequently lower profits for both, equal to 20 million SEK
- but if one cooperates (i.e., produces at low levels) and the other defects (i.e., surreptitiously produces at high levels), then the former just breaks even (i.e., no profits). At the same time, the latter earns a profit of 70 million SEK.

Represent the above situation as a game in its Normal Form. What is the dominant strategy of each company, and why? Can we predict what the outcome of this game will be? Do we make any assumptions to make this prediction?

Marking Notes:

We represent the game in its Normal Form, as shown in the payoff matrix below:

		STIHL	
		Cooperate	**Defect**
Husqvarna	**Cooperate**	**4**,**4**	**0**,**7**
	Defect	**7**,**0**	**2**,**2**

About dominance, we know the following:

- A **dominant strategy** is a strategy that **dominates all other strategies**.
- A strategy profile that consists of **dominant strategies for each player**, must necessarily constitute a **Nash Equilibrium** (no player wants to change strategy). Two types of dominance exist:
  - **Strict dominance:** For every strategy that the other players can play, if the utility of strategy s_i for player i is larger than any other strategy s'_i for player i, then there is a strict dominance of strategy s_i over strategy s'_i.
  - **Weak dominance:** The same as the above, but strategy s_i is larger or equal to strategy s'_i, which means that there is a weak dominance of strategy s_i over strategy s'_i.
- An **equilibrium that only consists of strict dominant strategies** must be **unique**.

For **Husqvarna**, strategy **Cooperate** is strictly dominated by strategy **Defect**, since:

- if $a_{STIHL} = Cooperate$ then $a^*_{Husqvarna} = Defect$, where $u_{Husqvarna}(Defect, Cooperate) = 7 > u_{Husqvarna}(Cooperate, Cooperate) = 4$.
- if $a_{STIHL} = Defect$ then $a^*_{Husqvarna} = Defect$, where $u_{Husqvarna}(Defect, Defect) = 2 > u_{Husqvarna}(Cooperate, Defect) = 0$.

For **STIHL**, strategy **Cooperate** is strictly dominated by strategy **Defect**, since:

- if $a_{Husqvarna} = Cooperate$ then $a^*_{STIHL} = Defect$, where $u_{STIHL}(Cooperate, Defect) = 7 > u_{STIHL}(Cooperate, Cooperate) = 4$.
- if $a_{Husqvarna} = Defect$ then $a^*_{STIHL} = Defect$, where $u_{STIHL}(Defect, Defect) = 2 > u_{STIHL}(Defect, Cooperate) = 0$.

Therefore, the dominant strategy for **Husqvarna** is **Defect**, and for **STIHL** is **Defect**, as well.

Considering that for an outcome that constitutes a **Nash Equilibrium**, **no agent can increase its utility** by changing to another action, we predict that the game's outcome will be (**Defect**,**Defect**).

To predict that outcome, we assumed that **both companies** would act **rationally**.

Marking:

- -0.5p for incorrect or missing Normal Form.
- -0.5p for each incorrect or missing dominant strategy. ($\times 2$)
- -0.5p for the incorrect or missing outcome of the game. ($\times 2$)
- -0.5p for incorrect or missing assumptions.

(c) (4 points) Calculate a Nash Equilibrium with mixed strategies from the reduced payoff matrix in (a) and state the value of the game at this equilibrium. Use the method of assuming the opponent should be indifferent between its actions. Show all your calculations.

Marking Notes:
The reduced payoff matrix is reiterated below:

		Player B	
		Y	**Z**
Player A	**X**	**4**,**6**	**1**,**5**
	Y	**3**,**3**	**2**,**6**

We can denote the probability that **Player A** plays strategy **X** with **p** (and **1-p** for strategy **Y**) and that **Player B** plays strategy **Y** with probability **q** (and **1-q** for strategy **Z**).

			Player B	
			q	**1-q**
			Y	**Z**
Player A	**p**	**X**	**4**,**6**	**1**, **5**
	1-p	**Y**	**3**,**3**	**2**,**6**

We then reason that **Player B** has chosen their probabilities **q** and **1-q** in such a way that **Player A** is indifferent between strategy **X** and **Y**. Likewise, we reason that **Player A** has chosen their probabilities **p** and **1-p** in such a way that **Player B** is indifferent between strategy **Y** and **Z**. This gives us the equations below:

Player A (row X = row Y)

$$\begin{aligned} 4q + 1(1-q) &= 3q + 2(1-q) \\ 4q + 1 - q &= 3q + 2 - 2q \\ 2q &= 1 \\ q &= 1/2 \\ 1 - q &= 1/2 \end{aligned}$$

Player B (column Y = column Z)

$$\begin{aligned} 6p + 3(1-p) &= 5p + 6(1-p) \\ 6p + 3 - 3p &= 5p + 6 - 6p \\ 4p &= 3 \\ p &= 3/4 \\ 1 - p &= 1/4 \end{aligned}$$

This gives us q = 1/2, 1-q = 1/2 and p = 3/4, 1-p = 1/4. Hence, we have a Nash Equilibrium with the mixed strategies:

- Player A plays strategy X 3/4 of the time and strategy Y 1/4 of the time.
- Player B plays strategy X 1/2 of the time and strategy Y 1/2 of the time.

The value of the game at this mixed-strategy Nash Equilibrium is obtained by multiplying each pay-off for each player in each outcome (cell) with the corresponding probability for Player A and Player B in that outcome, and finally summing the result in each cell (i.e., the expected value):

Player A

$$\begin{aligned} U_A &= 4pq + 1p(1-q) + 3(1-p)q + 2(1-p)(1-q) \\ &= 4pq + p - pq + 3q - 3pq + 2 - 2q - 2p + 2pq \\ &= 2pq - p + q + 2 \\ &= 2 \times 3/4 \times 1/2 - 3/4 + 1/2 + 2 \\ &= 3/4 - 3/4 + 1/2 + 2 \\ &= 1/2 + 4/2 \\ &= 5/2 \\ &= 2.50 \end{aligned}$$

Player B

$$\begin{aligned} U_B &= 6pq + 5p(1-q) + 3(1-p)q + 6(1-p)(1-q) \\ &= 6pq + 5p - 5pq + 3 - 3pq + 6 - 6q - 6p + 6pq \\ &= 4pq - p - 3q + 6 \\ &= 4 \times 3/4 \times 1/2 - 3/4 - 3 \times 1/2 + 6 \\ &= 3/2 - 3/4 - 3/2 + 24/4 \\ &= 21/4 \\ &= 5.25 \end{aligned}$$

The value of the game is, hence: $(U_A, U_B) = (2.50, 5.25)$.

Marking:

- -1p for an incorrect mixed strategy per player. (Mixed strategies: 2p)
- -1p for an incorrect game value per player. (Value of the game: 2p)

Total for Question 2: 12

3. MDP / Reinforcement Learning

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a discount factor $\gamma = 1/2$, a reward function R and a transition function T as shown below:

$$R(s, a, s') = \begin{cases} 1 & \text{if } s' = S_1 \\ -1 & \text{if } s' = S_2 \end{cases}$$

s	a	s'	T(s,a,s')
S_1	A_1	S_1	1/2
S_1	A_1	S_2	1/2
S_1	A_2	S_1	1
S_2	A_1	S_1	1/2
S_2	A_1	S_2	1/2
S_2	A_2	S_1	1/3
S_2	A_2	S_2	2/3

(a) (3 points) Perform one iteration of value iteration and enter the value for $V_1(S_1)$ in the table below (you don't have to enter a value into the gray cell). Use the initial values for $V_0(s)$ below instead of initializing all values to 0. Show all your calculations.

s	$V_0(s)$	$V_1(s)$
S_1	2	?
S_2	3	

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma V_k(s')\right]$$

Marking Notes:

By substituting values into the value iteration update equation we get:

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s, a, s') \left[R(s, a, s') + \gamma V_k(s')\right]$$

$$V_1(S_1) \leftarrow max \begin{pmatrix} A_1 : T(S_1, A_1, S_1)\left[R(S_1, A_1, S_1) + \gamma V_0(S_1)\right] + T(S_1, A_1, S_2)\left[R(S_1, A_1, S_2) + \gamma V_0(S_2)\right] \\ A_2 : T(S_1, A_2, S_1)\left[R(S_1, A_2, S_1) + \gamma V_0(S_1)\right] + T(S_1, A_2, S_2)\left[R(S_1, A_2, S_2) + \gamma V_0(S_2)\right] \end{pmatrix}$$

$$V_1(S_1) \leftarrow max \begin{pmatrix} A_1 : \frac{1}{2} * \left[1 + \frac{1}{2} * 2\right] + \frac{1}{2} * \left[-1 + \frac{1}{2} * 3\right] = 1 + \frac{1}{4} = \frac{5}{4} = 1.25 \\ A_2 : 1 * \left[1 + \frac{1}{2} * 2\right] + 0 * \left[-1 + \frac{1}{2} * 3\right] = 2 + 0 = 2.00 \end{pmatrix}$$

s	$V_0(s)$	$V_1(s)$
S_1	2	2.00
S_2	3	

Marking:

- -0.5p for each incorrect or missing parameter in the equation.

(b) (6 points) Assume Q-learning with a learning rate $\alpha = 1/2$ is used, and the following

episode is observed (i.e. first the agent executes the action A_1 in state S_1, ending up in state S_1 with the reward *1*, followed by the agent executing the action A_1 in state S_1, ending up in state S_2 with the reward *-1*):

s_1	a_1	r_1	s_2	a_2	r_2	s_3
S_1	A_1	1	S_1	A_1	-1	S_2

Use the initial Q-values Q_0 in the table below, and enter the values for Q_1 and Q_2. Show all your calculations.

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	-1/2	?	?
S_1	A_2	0	?	?
S_2	A_1	-1	?	?
S_2	A_2	1	?	?

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha \left[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a) \right]$$

Marking Notes:

By substituting values into the q-value update equation we get:

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha \left[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a) \right]$$

First step:

$$\begin{aligned} Q_1(S_1,A_1) &\leftarrow Q_0(S_1,A_1) + \frac{1}{2}\left[1 + \frac{1}{2}\max_{a'} Q_0(S_1,a') - Q_0(S_1,A_1)\right] \\ &\leftarrow -\frac{1}{2} + \frac{1}{2}\left[1 + \frac{1}{2}\max(-\frac{1}{2},0) - (-\frac{1}{2})\right] = -\frac{1}{2} + \frac{3}{4} = \frac{1}{4} \end{aligned}$$

$Q_1(S_1,A_2) \leftarrow Q_0(S_1,A_2)$ since A_2 was not taken in S_1 during step 1 in the episode
$Q_1(S_2,A_1) \leftarrow Q_0(S_2,A_1)$ since A_1 was not taken in S_2 during step 1 in the episode
$Q_1(S_2,A_2) \leftarrow Q_0(S_2,A_2)$ since A_2 was not taken in S_2 during step 1 in the episode

Second step:

$$\begin{aligned} Q_2(S_1,A_1) &\leftarrow Q_1(S_1,A_1) + \frac{1}{2}\left[-1 + \frac{1}{2}\max_{a'} Q_1(S_2,a') - Q_1(S_1,A_1)\right] \\ &\leftarrow \frac{1}{4} + \frac{1}{2}\left[-1 + \frac{1}{2}\max(-1,1) - \frac{1}{4}\right] = \frac{1}{4} - \frac{3}{8} = \frac{2}{8} - \frac{3}{8} = -\frac{1}{8} \end{aligned}$$

$Q_2(S_1,A_2) \leftarrow Q_1(S_1,A_2)$ since A_2 was not taken in S_1 during step 2 in the episode
$Q_2(S_2,A_1) \leftarrow Q_1(S_2,A_1)$ since A_1 was not taken in S_2 during step 2 in the episode
$Q_2(S_2,A_2) \leftarrow Q_1(S_2,A_2)$ since A_2 was not taken in S_2 during step 2 in the episode

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	1/2	1/4	-1/8
S_1	A_2	0	0	0
S_2	A_1	-1	-1	-1
S_2	A_2	1	1	1

Marking:

- -0.5p for each incorrect or missing parameter in the equation.

Total for Question 3: 9

4. Logic

(a) (7 points) First-order logic

Translate the following English sentences into first-order logic (FOL). Use the unary predicates *cat, dog, happy, falling, white*, the binary predicate *likes* and the constants *roddy, selma, snow, Christmas*. You must use exactly one FOL sentence per English sentence

1. All dogs like snow.
2. No cat likes snow.
3. A white Christmas make all dogs happy and all cats unhappy.
4. A white Christmas happens when snow is falling.
5. Selma is a cat.
6. Roddy is a dog.
7. Snow is falling.

Marking Notes:
Solution:

1. $\forall x$ dog(x) $\rightarrow$ likes(x, snow)
2. $\forall x$ cat(x) $\rightarrow \neg$ likes(x, snow) or $\neg\exists x$ cat(x) $\wedge$ likes(x, snow)
3. white(Christmas) $\rightarrow$ ($\forall x$ dog(x) $\rightarrow$ happy(x)) $\wedge$ ($\forall x$ cat(x) $\rightarrow \neg$ happy(x))
4. falling(snow) $\rightarrow$ white(Christmas)
5. cat(selma)
6. dog(roddy)
7. falling(snow)

Marking: 1 point is deducted for each mistake, such as using the wrong predicates, or failing to correctly capture the meaning of the English sentence.

(b) (7 points) Resolution proof

Transform your FOL sentences above into CNF and carry out a resolution proof of the sentence *Roddy is happy*. For full points, all sentences must be correctly re-written into CNF, including any that are not needed to the proof.

Marking Notes:
Solution, sentences in CNF:

- $\neg$ dog(x) $\vee$ likes(x, snow)
- $\neg$ cat(x) $\vee$ likes(x, snow)
- $\neg$ white(Christmas) $\vee$ $\neg$ dog(x) $\vee$ happy(x)
  $\neg$ white(Christmas) $\vee$ $\neg$ cat(x) $\vee$ $\neg$ happy(x)
- $\neg$ falling(snow) $\vee$ white(Christmas)
- cat(selma)
- dog(roddy)
- falling(snow)

Negation of conclusion is added to the knowledge base, i.e. $\neg$ happy(roddy), and the entire knowledge base is the following sentences;

1. $\neg$ dog(x) $\vee$ likes(x, snow)
2. $\neg$ cat(x) $\vee$ likes(x, snow)
3. $\neg$ white(Christmas) $\vee$ $\neg$ dog(x) $\vee$ happy(x)
4. $\neg$ white(Christmas) $\vee$ $\neg$ cat(x) $\vee$ $\neg$ happy(x)
5. $\neg$ falling(snow) $\vee$ white(Christmas)
6. cat(selma)
7. dog(roddy)
8. falling(snow)
9. $\neg$ happy(roddy)

The resolution proof then becomes:

10. white(Christmas) 5, 8
11. $\neg$ dog(x) $\vee$ happy(x) 10, 3
12. happy(roddy) 11, 7
13. $\square$ 12, 9

Marking:

- 3.5 point for correct conversion, with 0.5 deduction for each serious mistake.
- 3.5 point for the resolution proof.
- 0 points for all proofs that do not convert into CNF and/or does not only use resolution rule.

Total for Question 4: 14

5. Prolog

(a) (8 points) Cut

Write a predicate `split/3` that splits the list in the first argument into a list of positive (including zero) elements in the second argument and a list of negative elements in the third argument. For example:

```
?- split([1, -2, 3, 0], L1, L2).
L1 = [1, 3, 0],
L2 = [-2].
```

Marking Notes:

For a sample solution, see Exercise 3, Task 4.
4 points are awarded for correct splittin of the list.
4 points are awarded for the use of the cut.
0 point is awarded if the program does not work.
2 point is withdrawn if unnecessary clauses are used in the program.

(b) (8 points) CLP.

Find all the solutions for the cryptarithmetic puzzle below using CLP(FD).

The cryptarithmetic puzzle

```
      CP
+     IS
+    FUN
--------
=   TRUE
```

Marking Notes:

To solve this problem, you need first to create an equation derived from the task and then use the label predicate to find concrete solutions. An arithmetic constraint should be written including each digit and depending on the digit ordinal place in a number.

You may want to revise the cryptarithmetic example from the course, to understand these solution instructions completely.

Marking: 5p are awarded for a correct solution of the puzzle with CLP(FD).
1p is awarded if the variables are correctly set up.
1p is awarded if the equation is correct.
1p is awarded for the correct outputs.
0p is given if the solution was incorrect.

(c) (7 points) Data Structures

Write a procedure `preorder_tree/2` that collects all the nodes in a binary tree using pre-order traversal: root, left subtree, right subtree. Use the atom `nil` to represent an empty tree, and the term `t(L,X,R)` to represent a non-empty binary tree where `X` denotes the root node and `L` and `R` denote the left and right subtree, respectively.

Marking Notes:
It is a recursive rule. To solve this problem, the important part is that you understand what a binary tree is and how it is represented in Prolog as required. For a sketch of a solution, see the similar tasks (7 and 8) in Exercise 3.
0p is given if the program does not work.
2p is withdrawn for unnecessary clauses/cuts in the program, or if the program is unnecessarily complex.

Total for Question 5: 23